

# Harikeshav Rameshkumar

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## Education

**The Ohio State University** — Dean's List & University Honors (all semesters)  
*B.S. Computer Science (Accelerated 3-Year Track); GPA: 3.9/4.0*

**Columbus, OH**  
*Expected May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Systems Programming, Computer Networking, Principles of Programming Languages, Database Systems, Software Engineering, Full Stack Web Development, Deep Learning & AI

## Technical Skills

**Languages:** Python, C++, TypeScript, JavaScript, Rust, C, C#, SQL

**AI / ML & LLMs:** Multi-Agent Systems (**LangChain**, **LangGraph**), LLM Integration (**Anthropic Claude**, **OpenAI GPT-4o**, **Google Gemini**), **AWS Bedrock**, **RAG**, Fine-Tuning (BERT / Transformers, PyTorch), TensorFlow, Prompt Engineering

**Systems & Security:** Distributed Systems, SIMD (AVX2 / NEON), Linux Kernel Modules, Multithreading & Concurrency, Compilers / IR, Quantization, Cryptography (FHE, AES, X3DH / Double Ratchet), TCP / Sockets

**Backend & Web:** **FastAPI**, Flask, **React**, Node.js, REST APIs, WebSockets, Microservices

**Cloud & DevOps:** **AWS** (Bedrock, ECS, Lambda, EKS, SageMaker), **Docker**, Kubernetes, **CI/CD** (GitHub Actions), pytest, mypy

**Databases:** **PostgreSQL**, MySQL, Redis, Vector DBs (Qdrant, ChromaDB, pgvector), Graph DBs (Amazon Neptune), SQLite

## Experience

**GE Aerospace**

**Bengaluru, India (Remote)**

*Software Engineering Intern, AI/FinFlow Team*

*Jun. 2026 – Aug. 2026*

- Worked on **FinFlow**, a production platform that extracts business rules, pricing, and milestone data out of legacy contracts and purchase orders; built the extraction pipeline on **AWS Bedrock + Claude**, cutting per-document latency **74%** (1,957s → 505s) through request streaming and rate-limiting.
- Engineered an **AI assistant** that lets finance users query a financial-lineage **graph database** in plain English instead of writing code (**FastAPI + Amazon Neptune**); hardened it for production — `mypy --strict` errors 93 → 0 across 31 modules, removed an `exec()` RCE vulnerability, and grew the `pytest` suite from 29 to **78 tests**.
- Extended the pipeline past Claude's 100-page request limit to handle 448-page (**\$101M**) contracts via a **page-split-and-merge algorithm** with content-hash caching, cutting API calls **55%** and runtime **37%**.
- Architected an adoption-analytics pipeline (Python, **Polars**, **DuckDB**) processing **726K+** logs across 100K+ employees in **15s**, and traced a data-integrity defect that had inflated the headline number, correcting the figure leadership presented.

**Siage Solutions**

**Bangalore, India**

*Software Engineering Intern*

*Jun. 2025 – Aug. 2025*

- Developed an internal knowledge assistant that lets engineers ask questions in plain English across 100+ technical manuals (15M+ tokens) instead of searching PDFs, using **RAG (LangChain, Qdrant, OpenAI embeddings)**; cut search latency **91%** (4.2s → 350ms) and became the team's daily documentation tool.
- Designed an automated ticket-triage engine that classifies and routes enterprise support tickets, fine-tuning a **BERT** classifier (PyTorch) against a vector index to cut duplicate tickets **42%** and accelerate Mean Time to Resolution to **4.0 hrs** (from 14.2).
- Deployed the real-time sentiment service behind these tools as a **vLLM** inference microservice (**AWS EKS**, Kubernetes, Docker) streaming to 2,500+ daily users, boosting token throughput **4.8×**.
- Scaled a distributed data-ingestion engine (Python **AsyncIO**, Playwright, **Celery**, **Redis**) across a 15-node proxy cluster, collecting **50,000+** reviews/day at **1,200 req/min**.

**Indian Institute of Technology, Madras**

**Remote**

*Research Intern, Wireless Networks & Spatial AI Group*

*Jun. 2023 – Oct. 2023*

- Modeled indoor Wi-Fi signal strength at any point in a building without physical site surveys, benchmarking 6 ML models (GPR, SVR, CNN, LSTM) in **Python/TensorFlow** to reach  $R^2 = 0.975$  accuracy on a synthetic 100,000+point dataset and cut manual site-survey effort **86%**.

## Projects

**LEAP** | *Distributed LLM Inference Engine* – C++20, SIMD, Linux Kernel, PyTorch

- A **distributed inference engine** that runs models too large for one GPU (e.g. Llama 3 70B) by splitting transformer layers across a ring of consumer machines using **pipeline parallelism**, bypassing the VRAM wall.
- Wrote hand-tuned **SIMD** kernels (**AVX2/FMA**, **ARM NEON**) with OpenMP, a zero-copy **Linux kernel module** (Netfilter, `mmap`) to bypass the network stack, and an **INT8** quantizer shrinking model memory **4×**.

**Penumbra-FHE** | *FHE Compiler for Encrypted ML Inference* – Python, Rust, ONNX, pytest

- An open-source **compiler + runtime** that runs neural-network inference directly on **encrypted data** via **Fully Homomorphic Encryption**, so a server computes predictions without ever seeing the plaintext.
- Built the front end as a **parser** that lowers **ONNX** compute graphs into a schema-versioned **IR** (Python → **Rust tfhe-rs** runtime), gated by a **behavior-equivalence proof** (encrypted output  $\equiv$  cleartext, *bit-for-bit*) verified in **CI/CD**.

**Sane-Jtreed** | *AI Agent-Simulated Hedge Fund* – LangGraph, ChromaDB, FastAPI, React

- A **multi-agent** platform that simulates a quantitative **hedge-fund research desk**: AI agents research a stock, debate bull vs. bear theses, and a **risk committee** approves or rejects the trade.
- Orchestrated agents across **Claude**, **GPT-4o**, and **Gemini** through a **LangGraph** state machine with **ChromaDB** long-term memory; shipped a full-stack dashboard (**React**, **FastAPI**, **PostgreSQL**, Redis).

**ArbOS** | *Prediction-Market Logical Arbitrage Engine* – Rust, Tokio, watsonx.ai, LangGraph

- A low-latency **Rust** engine that performs **logical arbitrage** on decentralized prediction markets: it watches up to **1,000** live **Polymarket** order books and, the instant prices violate the axioms of probability, fires simultaneous **delta-neutral** orders (guaranteed at resolution) at **~14 ms** detection.
- Engineered an **LLM reasoning “Brain”** (**IBM watsonx.ai**, **LangGraph**, **NetworkX**) that autonomously maps the hidden logical graph linking markets to drive **4** arbitrage strategies, hardened by **Fill-or-Kill**-only execution, circuit breakers, and exact `rust_decimal` fixed-point money math.

## Awards

- **2nd Place, Medpace Track, RevolutionUC 2026** (60+ teams) — *Pulse*, multi-agent clinical-trial safety platform
- **3rd Place, Visa Track, TartanHacks 2026** (Carnegie Mellon, 1,200+ hackers) — *Penny*, AI personal-finance assistant
- **3rd Place, The Token Company Track, NextHacks 2026** (2,000+ participants) — *Distill*, LLM context-compression engine
- **Best AI Hack Runner-Up, HackOHI/O 2025** (800+ participants) — *LeadForge*, autonomous agentic AI sales platform